Fixed-axis core collection after a vortex-phase transformation of a Gaussian beam

Auto BeamScientist Project

Abstract

A phase-only transformation preserves an input beam's intensity while changing the phase that governs its subsequent diffraction. We test a registered, condition-specific prediction: for a Gaussian beam with a 0.10 mm amplitude waist, applying a charge +2 vortex phase in free space gives a transformed-to-base ratio no greater than one for the median power fraction in a fixed on-axis circular core over $z = 0.03$–0.08 m. The 633 nm, 1 W fields are propagated without a lens, and the core radius is 0.10 mm. At the reference numerical tier, the transformed and base median core efficiencies are 0.01763695 and 0.59284526, respectively, giving a ratio of 0.02974967. The transformed core fraction is lower at all five sampled planes. A charge-zero identity control passes; mesh, window, axial-grid, wavelength, quantization, noise-probe, and independent-propagator checks reproduce the directional comparison across the tested variants. These results support the registered prediction for this scalar finite-core observable. They do not isolate a vortex-specific mechanism, establish a charge law or topology, or establish novelty.

1 Introduction

Power collected within a specified aperture is not interchangeable with peak intensity or with the position of the brightest point. In an on-axis collection geometry, a direct measure is the fraction of propagated power within a fixed region about the optical axis. A phase-only input transformation leaves the input intensity unchanged, but changes the phase structure on which diffraction acts and can therefore alter this collected fraction.

Vortex phases have been studied in the context of structured light and spiral phase plates. Prior work addresses diffraction of Gaussian beams by a spiral phase plate [1]. The present question is more narrowly defined by its observable and comparison: for one registered Gaussian waist and charge, does the transformed field collect less power in a fixed on-axis core, summarized by the median over a specified free-space interval? The endpoint is an integrated-intensity statistic, not a measurement of modal content, orbital angular momentum, or vortex topology.

We preserve the registered question: for a Gaussian input with a 0.10 mm waist in free space, does a charge +2 vortex-phase transformation produce a transformed-to-base median fixed-axis core-efficiency ratio no greater than one over 0.03–0.08 m? The primary comparison uses matched transformed and untransformed arms, with a charge-zero identity audit as a separate implementation control. The results support the stated direction for this configuration. Since no matched non-vortex phase intervention was evaluated, they do not establish that the collection change is specific to vortex phase rather than to phase modulation more generally.

2 Related Work

Kotlyar *et al.* report work on diffraction of conic and Gaussian beams by a spiral phase plate [1]. This is relevant to the broad operation considered here: imposing a spiral phase and studying subsequent diffraction. The present numerical comparison uses a Gaussian input and a charge +2 phase, but focuses on a particular finite-core quantity: the fraction of input power inside a fixed

on-axis circle, summarized over a prescribed propagation interval. The verified bibliographic record establishes the topic of the cited work, but its full text has not been reviewed here. We therefore make no detailed claims about overlap or differences in methods, conditions, or conclusions.

The observable used below characterizes collection in the specified geometry. It does not, by itself, diagnose a vortex, establish a topological charge, or identify a unique cause for the change in collection. The available prior-art information is not sufficient to establish a novelty gap, and no novelty claim is made.

3 Theoretical model

3.1 Input field and vortex transformation

Let $V(x, y)$ be the sampled, unnormalized input field. The Gaussian source used here is

$$V(x,y) = \begin{cases} \exp[-(x^2+y^2)/w^2], & x^2+y^2 \le R_{\mathrm{ap}}^2, \\ 0, & x^2+y^2 > R_{\mathrm{ap}}^2, \end{cases} \tag{1}$$

where $w = 0.10$ mm is the amplitude waist and $R_{\mathrm{ap}} = 0.9$ mm is the physical aperture radius. The sampled field is normalized to input power P:

$$U_0(x_j, y_k) = \sqrt{\frac{P}{\Delta x^2 \sum_{a,b} |V_{ab}|^2}}\, V(x_j, y_k). \tag{2}$$

Here $\Delta x = \Delta y$ is the transverse sample spacing, and the primary setup uses $P = 1$ W.

The transformation applied to the primary arm is a vortex phase,

$$V_m(x,y) = V(x,y) \exp[im \operatorname{atan2}(y,x)], \tag{3}$$

with $m = +2$. The comparison base is the same sampled Gaussian without this phase factor. Thus, up to numerical precision, the transformation preserves the input amplitude while changing the phase. At the grid origin, NumPy assigns $\operatorname{atan2}(0,0) = 0$; the phase factor at that pixel is therefore 1, with no amplitude regularization. This specifies the finite-grid convention; it does not resolve the continuum phase singularity or establish a topological property. The finite sampled grid and its origin convention are part of the numerical model.

For clarity, the general coherent composition convention is

$$U_c = \sqrt{\frac{P}{\int |U + a e^{i\delta} \mathcal{R}_\theta U|^2 \, dA}} \left(U + a e^{i\delta} \mathcal{R}_\theta U \right), \tag{4}$$

where $\mathcal{R}_\theta$ denotes rotation by θ, a is the relative amplitude weight, and δ is the relative phase. The executed comparison uses $\theta = 0$, $\delta = 0$, and $a = 1$, but it is a single transformed arm compared with a single base arm, not a coherent two-arm rotational superposition. If a rotation is applied, the real and imaginary fields are sampled independently by bilinear interpolation, with constant-zero boundaries around the physical grid origin, and the rotated arm is normalized to input power. An exactly zero coherent field cannot be normalized.

3.2 Propagation and core-efficiency observable

Free-space propagation is calculated with the angular-spectrum method (ASM):

$$U_z = \mathcal{F}^{-1}\left[\mathcal{F}(U_0) B_z \exp\left(i 2\pi z \sqrt{\lambda^{-2} - f_x^2 - f_y^2} \right) \right], \tag{5}$$

where $\mathcal{F}$ is the transverse Fourier transform, f_x and f_y are spatial frequencies, $\lambda = 633$ nm, and B_z is the numerical bandwidth mask. The conservative radial cutoff used in the calculation is

$$\nu_c(z) = \min\left[\lambda^{-1}, \frac{\lambda^{-1}}{\sqrt{1 + (2z/L)^2}}\right], \tag{6}$$

where L is the computational-window width. This is a conservative radial cutoff, not the complete Matsushima passband construction. The executed setup applies no lens.

For a fixed-axis circular core of radius $r_c = 0.10$ mm, the core efficiency at plane z and its median over the sampled planes are

$$C(z) = P^{-1} \int_{x^2+y^2 \le r_c^2} |U_z|^2 \, dA, \qquad \overline{C} = \mathrm{median}_{z \in \mathcal{Z}} \, C(z). \tag{7}$$

The primary observable is the transformed-to-base ratio of these arm-wise medians:

$$R_{\mathrm{core}} = \frac{\mathrm{median}_{z \in \mathcal{Z}} \, C_{\mathrm{vortex}}(z)}{\mathrm{median}_{z \in \mathcal{Z}} \, C_{\mathrm{base}}(z)}. \tag{8}$$

The registered prediction is $R_{\mathrm{core}} \le 1$. In the reported calculation, the core integral is approximated by summing intensities of pixels whose centers satisfy $r \le r_c$, then multiplying by Δx^2. It is therefore a pixel-center approximation to the circular boundary.

4 Numerical methods

The reference calculation uses $N = 2048$ samples per transverse dimension on a square window of width $L = 8$ mm. Grid coordinates are

$$x_j = (j - \lfloor N/2 \rfloor)L/N, \qquad y_k = (k - \lfloor N/2 \rfloor)L/N.$$

The executed setup has wavelength 633 nm, amplitude waist 0.10 mm, input power 1 W, and the 0.9 mm physical aperture. Its recorded focal-length parameter is 0.06 m, but a lens is not applied. ASM propagation is evaluated at

$$z = (0.030,\ 0.0425,\ 0.055,\ 0.0675,\ 0.080) \text{ m}.$$

The transformed and base arms share the source, aperture, input-power convention, grid, propagation environment, axial planes, and fixed-axis core.

The primary comparison passed the recorded finite-field, equal-power, sampling-retention, and window-edge checks. The minimum retained output power for the transformed arm at the reference tier was 0.9970418 W, and its maximum recorded edge fraction was 0.001515. The charge-zero audit was evaluated as a separate identity control. Its transformed-to-base relative field difference is zero in the primary record, and confirmation validates the identity within a pre-established relative-L^2 tolerance of 10^{-10}. This is an implementation control, not a nonzero physical finding.

Numerical checks include mesh refinement, changes in computational window, denser axial sampling, an independent-observable check, and comparison with a paraxial Fresnel propagator. Quantization, a held-out wavelength, and two noise probes were also evaluated. These variants test numerical stability or robustness; they are not independent physical samples. The noise probes are not required to preserve an identity comparison. The primary core quadrature uses pixel-center inclusion, and the supplied records do not isolate a separate error attributable only to discretization of the circular boundary. The reported numerical checks are therefore validation of the comparison under the tested implementations, not a complete uncertainty estimate.

5 Results and physical interpretation

5.1 Primary comparison

At the reference numerical tier, the transformed and base arm medians are

$$\overline{C}_{\mathrm{vortex}} = 0.01763695, \qquad \overline{C}_{\mathrm{base}} = 0.59284526.$$

Their ratio is $R_{\mathrm{core}} = 0.02974967$, below the registered threshold of one. Thus, the directional prediction is supported for the specified scalar field, propagation interval, and collection core. The ratio is calculated from the two arm-wise medians; it is not a ratio of incident-power throughput or an absolute measure of concentration.

Table 1 reports the saved plane-wise core fractions. Both arms decrease monotonically with increasing propagation distance at the five sampled planes, and the transformed arm has lower core efficiency than the base at each plane. The median comparison therefore reflects a consistent arm ordering at these sampled positions.

Table 1: Reference-tier fixed-axis core fractions for the charge +2 and base Gaussian arms. Each value is the fraction of the arm's normalized input power within the 0.10 mm-radius core.

z (m)	Charge +2	Base Gaussian
0.0300	0.116445	0.769228
0.0425	0.042899	0.684965
0.0550	0.017637	0.592845
0.0675	0.007878	0.504676
0.0800	0.003780	0.426729
Median	0.017637	0.592845

The charge-zero control returns identical base and transformed median efficiencies, 0.59284526, with zero relative field difference in the primary record. Its confirmed identity demonstrates that the zero-charge operation leaves this sampled field unchanged within the stated tolerance. It does not constitute evidence for the charge +2 prediction; that inference comes from the separate primary comparison.

5.2 Numerical checks and scope of the inference

The comparison is stable across the reported numerical variants. The reference result uses $N = 2048$ and $L = 8$ mm. At the mesh-check tier ($N = 1536$, $L = 8$ mm), the transformed and base medians are 0.01739957 and 0.59081154, giving a ratio of 0.02945029. The window check ($N = 2304$, $L = 9$ mm) gives medians 0.01763707 and 0.59284526, with ratio 0.02974987. A second-window check ($N = 2560$, $L = 10$ mm) gives 0.01763718 and 0.59284526, with ratio 0.02975005. These values show that the paired directional comparison is stable across these saved mesh and window variants, while documenting that the absolute arm medians vary between numerical tiers.

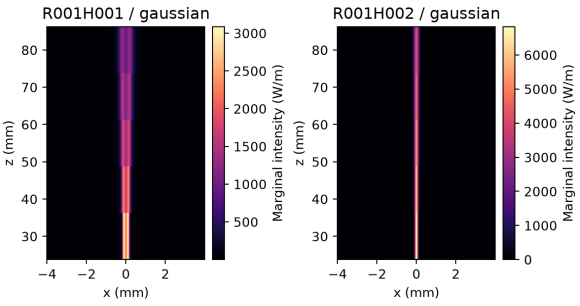
With the reference grid and window, nine axial planes give a ratio of 0.02974967, the same to the reported precision as the five-plane result. The independent paraxial Fresnel calculation gives transformed and base medians of 0.01763740 and 0.59284603, and a ratio of 0.02975038. The held-out wavelength of 640 nm gives 0.01700306 and 0.58839069, with ratio 0.02889756. The quantized-phase check gives a ratio of 0.02974830; the two noise probes give 0.02972996 and 0.02972573. These are numerical or robustness variants of the comparison, not additional physical replicates. The confirmation procedure's reported numerical error floor of 0.01 is a criterion used by that procedure, not a measured uncertainty interval for the core efficiencies.

The physical mechanism represented in the model is the effect of applying a different input phase before scalar diffraction: the phase changes while the input amplitude is preserved, and the propagated intensity distribution differs from that of the base. The measured quantity establishes that, under the stated conditions, less of the normalized transformed-arm power lies within the chosen fixed-axis core. No matched non-vortex phase control was run, so the comparison cannot determine whether the outcome is specific to the vortex transformation or shared by other phase-only changes. The core-efficiency statistic also does not measure topology.

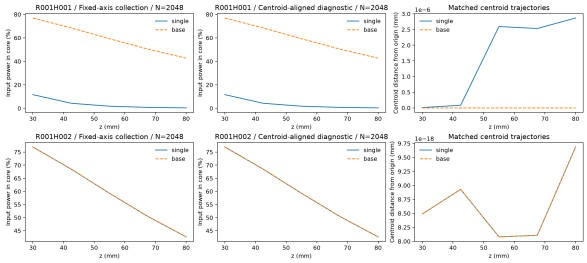
Fixed-axis collection should be distinguished from recentered collection. In the reference diagnostics, the charge +2 arm's centroid remains close to the axis; its largest recorded centroid displacement is about 2.9 nm. Its centroid-aligned core fractions are correspondingly nearly equal to its fixed-axis values. The base centroid is effectively on axis. Thus recentering does not materially change the reported core comparison in this calculation. Brightest-pixel alignment is a different diagnostic and is not the registered observable.

5.3 Figures

Figure 1 groups the saved propagation view with the matched axial diagnostics. The propagation view supplies visual context, while the matched diagnostics show the recorded axial comparison and alignment information. Table 1 gives the numerical values used for the primary inference. Figure 2 is supplementary and does not replace the arm-wise numerical report.



(a) Propagation view for the charge +2 case.



(b) Matched axial and alignment diagnostics.

Figure 1: Propagation context and matched diagnostics for the registered 0.10 mm-waist comparison. The primary statistic is the ratio of the two arm-wise median fixed-axis core efficiencies.

6 Conclusions

For a 0.10 mm-waist Gaussian input at 633 nm and 1 W, propagated in free space without a lens, the charge +2 vortex-phase arm has a median fixed-axis core efficiency of 0.01763695, compared with 0.59284526 for the matched base. The transformed-to-base median ratio is 0.02974967, supporting the registered prediction $R_{\mathrm{core}} \leq 1$ for the 0.10 mm-radius core over 0.03–0.08 m. The charge-zero identity control passes, and the numerical comparison is reproduced across the reported checks.

This conclusion is confined to the specified scalar field, finite grid, propagation interval, and collection geometry. The result does not establish a vortex-specific cause, a general charge dependence, or a topological property. A matched non-vortex phase control would be needed to distinguish the vortex intervention from broader phase-induced redistribution; the available prior-art evidence likewise does not support a novelty claim.

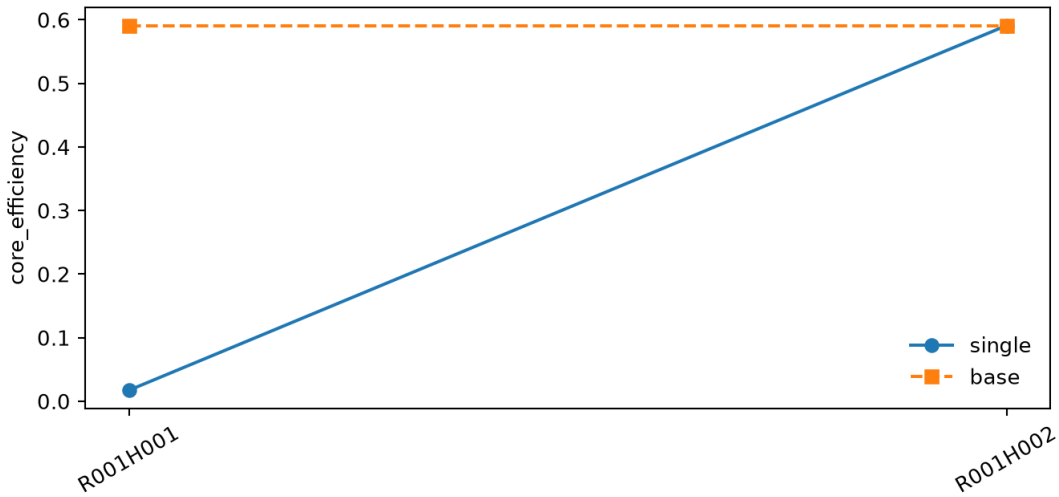


Figure 2: Saved overview of numerical results. Quantitative primary arm values are reported in Table 1, and numerical-tier comparisons are reported in the text.

Acknowledgement

Auto BeamScientist adapts Sakana AI's AI Scientist-v2 research workflow [2].

References

[1] Victor V. Kotlyar, Alexey A. Kovalev, Svetlana N. Khonina, Roman V. Skidanov, Victor A. Soifer, Henna Elfstrom, Noora Tossavainen, and Jari Turunen. Diffraction of conic and gaussian beams by a spiral phase plate. *Applied Optics*, 45(12):2656, 2006.

[2] Sakana AI. The ai scientist-v2. `https://github.com/SakanaAI/AI-Scientist-v2`.